1.Aim of the practical: The aim of this project is to convert and deploy the Employee Data Application using Enterprise Java Beans (EJB) while ensuring data integrity through Bean Validation. Additionally, the project focuses on implementing a one-to-one entity relationship mapping to establish proper database relationships and improve data consistency.

2.Objective: The objective of this project is to develop an enterprise-level Employee Data Application using EJB and apply Bean Validation to enforce data constraints. It also aims to implement a one-to-one entity relationship mapping using Java Persistence API (JPA) to efficiently manage database relationships. Through this, the project ensures scalability, reliability, and maintainability of the application.

3.Algorithm/Flowchart:

A) HTML CODE :

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Welcome</title>

    <!-- ✅ Bootstrap CDN -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
    <style>
        body {
            background: linear-gradient(135deg, #6a11cb, #2575fc);
            height: 100vh;
            display: flex;
```

```
            align-items: center;
            justify-content: center;
            color: #fff;
        }
        .welcome-card {
            padding: 40px;
            border-radius: 15px;
            background: rgba(255, 255, 255, 0.2);
            backdrop-filter: blur(10px);
            box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);
            text-align: center;
            max-width: 400px;
        }
        .welcome-card h2 {
            font-weight: bold;
        }
        .btn-custom {
            padding: 12px 20px;
            font-weight: bold;
            border: none;
            transition: 0.3s;
        }
        .btn-primary {
            background: #ff7eb3;
        }
        .btn-primary:hover {
            background: #ff5277;
        }
        .btn-success {
            background: #48c78e;
        }
        .btn-success:hover {
            background: #36a572;
        }
    </style>
</head>
<body>

    <div class="welcome-card">
        <h2>Welcome to User Authentication System</h2>
        <p class="text-light">Secure and reliable login system</p>

        <div class="mt-4 d-grid gap-3">
            <a href="register.jsp" class="btn btn-primary btn-custom">Register</a>
            <a href="login.jsp" class="btn btn-success btn-custom">Login</a>
        </div>
    </div>

    <!-- ✅ Bootstrap JS -->
```

```
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Hibernate Code:

```
<?xml version="1.0" encoding="UTF-8"?>
<hibernate-configuration>
    <session-factory>
        <!-- JDBC Database connection settings -->
        <property name="hibernate.dialect">org.hibernate.dialect.MySQL8Dialect</property>
        <property name="hibernate.connection.driver_class">com.mysql.cj.jdbc.Driver</property>
        <property name="hibernate.connection.url">jdbc:mysql://localhost:3306/DBemployee</property>
        <property name="hibernate.connection.username">root</property>
        <property name="hibernate.connection.password">Nikhil@123</property>

        <!-- JDBC connection pool settings -->
        <property name="hibernate.c3p0.min_size">5</property>
        <property name="hibernate.c3p0.max_size">20</property>
        <property name="hibernate.c3p0.timeout">300</property>

        <!-- Enable Hibernate's automatic session context management -->
        <property name="hibernate.current_session_context_class">thread</property>

        <!-- Echo all executed SQL to stdout -->
        <property name="hibernate.show_sql">true</property>

        <!-- Drop and re-create the database schema on startup -->
        <property name="hibernate.hbm2ddl.auto">update</property>

        <!-- Mention annotated class -->
        <mapping class="com.employee.model.Employee"/>
    </session-factory>
</hibernate-configuration>
```

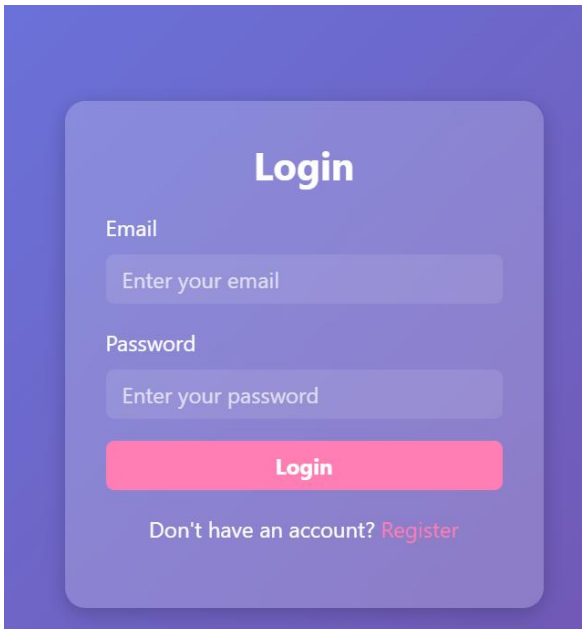
POM.xml:

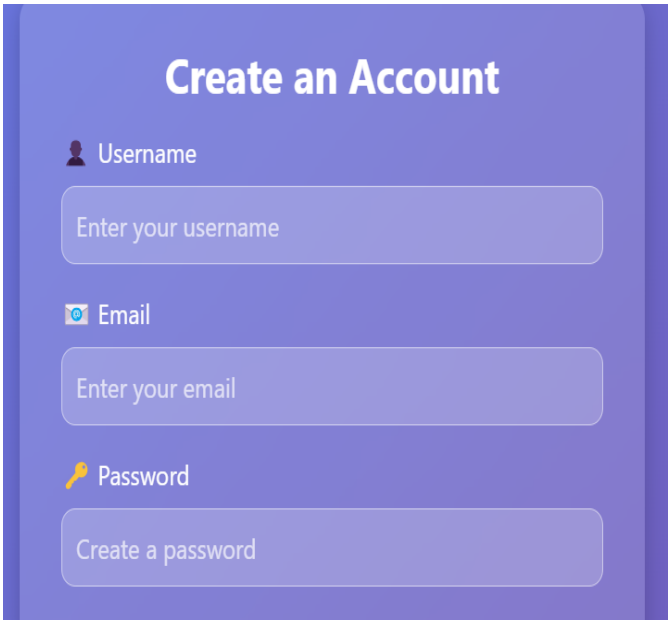
```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.employee</groupId>
    <artifactId>UserAuthSystem</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>war</packaging>
    <name>broooooo</name>

    <properties>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
```

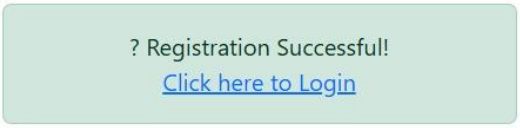
```
        <jakartaee>10.0.0</jakartaee>
    </properties>

    <dependencies>
        <dependency>
            <groupId>jakarta.platform</groupId>
            <artifactId>jakarta.jakartaee-api</artifactId>
            <version>${jakartaee}</version>
            <scope>provided</scope>
        </dependency>


        <dependency>
    <groupId>jakarta.servlet</groupId>
    <artifactId>jakarta.servlet-api</artifactId>
    <version>5.0.0</version>
    <scope>provided</scope>
</dependency>



        <dependency>
    <groupId>org.hibernate</groupId>
    <artifactId>hibernate-core</artifactId>
    <version>6.3.1.Final</version>
</dependency>


          <!-- MySQL JDBC Driver -->
        <dependency>
        <groupId>mysql</groupId>
        <artifactId>mysql-connector-java</artifactId>
        <version>8.0.33</version>
        <scope>runtime</scope>
        </dependency>


    </dependencies>



    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.8.1</version>
                <configuration>
                    <source>11</source>
```

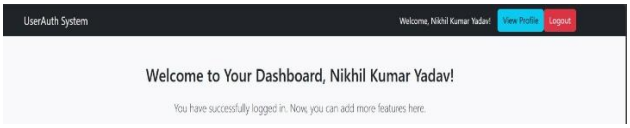
```
            <target>11</target>
          </configuration>
        </plugin>
        <plugin>
          <groupId>org.apache.maven.plugins</groupId>
          <artifactId>maven-war-plugin</artifactId>
          <version>3.3.2</version>
        </plugin>
      </plugins>
    </build>
</project>
```

4.Result/Output:









Learning outcomes (What I have learnt):

- Understand and implement Enterprise Java Beans (EJB) for enterprise applications.
- Apply Bean Validation to enforce data constraints and ensure accuracy.
- Implement one-to-one entity relationship mapping in Java Persistence API (JPA).
- Gain hands-on experience in deploying and managing enterprise applications.